ASCII MASTER FLOW — PANEL 1/12: Bit and qubit card

CLASSICAL BIT -> recorded as 0 or 1
QUBIT -> amplitudes over basis states
MEASUREMENT -> one classical outcome with probability
SPEED -> not the defining difference
RULE -> state description is not two stored classical bits

ASCII MASTER FLOW — PANEL 2/12: Quantum principle split

SUPERPOSITION -> one system in a combination of basis states
ENTANGLEMENT -> joint state of two or more systems
NO-CLONING -> arbitrary unknown state cannot be copied perfectly
NO SIGNALLING -> entanglement does not transmit messages faster
RULE -> each principle answers a different question

ASCII MASTER FLOW — PANEL 3/12: Algorithm mechanism rail

PREPARE STATE -> initialise qubits
QUANTUM GATES -> transform amplitudes
INTERFERENCE -> suppress and reinforce paths
MEASUREMENT -> obtain classical sample
TRAP -> superposition does not reveal every answer

ASCII MASTER FLOW — PANEL 4/12: Decoherence engineering stack

```
ENVIRONMENTAL INTERACTION -> phase information degrades
COHERENCE TIME -> finite computation window
CRYOGENICS / VACUUM / SHIELDING -> reduce disturbance
CONTROL ELECTRONICS / MATERIALS -> execute accurate gates
OUTCOME -> engineering, not theory alone, limits scale
```

ASCII MASTER FLOW — PANEL 5/12: Physical-to-logical ladder

```
PHYSICAL QUBIT -> noisy hardware element
ERROR CHARACTERISATION -> measure failure channels
ERROR-CORRECTING CODE -> redundant encoding
LOGICAL QUBIT -> protected information unit
RULE -> raw qubit count is not useful capability
```

ASCII MASTER FLOW — PANEL 6/12: Four technology families

COMPUTING -> selected information-processing algorithms
COMMUNICATION -> quantum-state links and key distribution
SENSING / METROLOGY -> precision measurement
MATERIALS / DEVICES -> enabling hardware substrate
TRAP -> quantum technology is not computing alone

ASCII MASTER FLOW — PANEL 7/12: QKD security chain

```
QUANTUM STATES -> distribute key material
DISTURBANCE TEST -> reveal possible interception
AUTHENTICATED ENDPOINTS -> prevent impersonation
CLASSICAL ENCRYPTION -> protects message content
RULE -> QKD alone is not a complete secure network
```

ASCII MASTER FLOW — PANEL 8/12: QKD versus PQC

```
QKD -> quantum hardware and physical link
PQC -> classical quantum-resistant algorithms
QKD PURPOSE -> specialised key distribution
PQC PURPOSE -> broad standards and software migration
POLICY -> complementary tools with different timelines
```

ASCII MASTER FLOW — PANEL 9/12: NQM institution map

```
MINISTRY OF SCIENCE AND TECHNOLOGY -> ministerial umbrella
DST -> implementing department
NQM -> national capability mission
T-HUBS -> four vertical ecosystem anchors
RULE -> department, mission and hub are distinct
```

ASCII MASTER FLOW — PANEL 10/12: Four T-Hub map

IISc BENGALURU -> Quantum Computing
IIT MADRAS WITH C-DOT -> Quantum Communication
IIT BOMBAY -> Quantum Sensing and Metrology
IIT DELHI -> Quantum Materials and Devices
TRAP -> hub creation is not operational capability
MUST REMEMBER: Quantum technology covers computing, communication and sensing: qubits...

ASCII MASTER FLOW — PANEL 11/12: Readiness ladder

CABINET APPROVAL -> mission authority and outlay
TARGET -> intended future capability
RESEARCH PROTOTYPE -> bounded demonstration
VALIDATED SYSTEM -> repeatable performance evidence
OPERATIONAL DEPLOYMENT -> field use with sustainment
CLOSE DISTINCTION: Qubit is not a faster classical bit, gate count is not useful...

ASCII MASTER FLOW — PANEL 12/12: Quantum claim firewall

```
DEFINE -> qubit, superposition, entanglement and decoherence
CLASSIFY -> computing, communication, sensing or materials
SEPARATE -> physical and logical qubits; QKD and PQC
LABEL -> approval, target, prototype, validation and operation
VERIFY -> outlay, qubits, distance and startup claims officially
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Identify platform, qubit/measurement...
```